Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694 · 089 632 5829
www.solaress.org · www.โซลาร์เอสส์.com

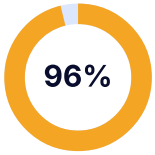
PLANT PERFORMANCE REPORT

Chicken farm, Chiang Rai

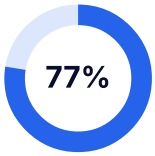
20 kWp grid-tied rooftop system · 20 kW inverter · no battery · Chiang Rai

PERIOD	DATA SOURCE	ISSUED
1 – 30 October 2026 (30 days)	Plant monitoring platform	31 October 2026

Solar production	Average per day	Farm consumption	Bought from grid	Sent to grid	Specific yield
1,458.0 kWh	48.6 kWh/day	1,807 kWh	407.8 kWh	58.5 kWh	2.43 kWh/kWp/day



of solar used on the farm

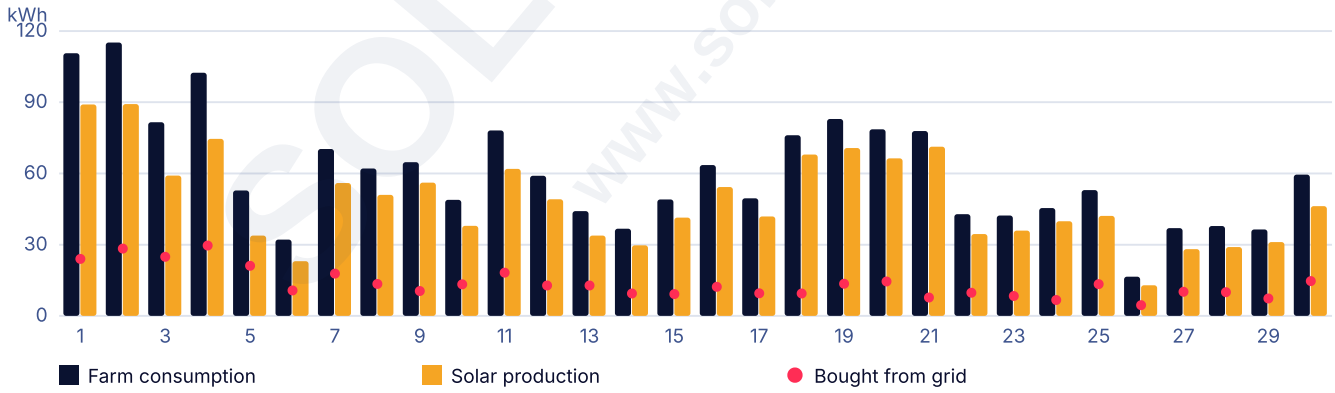


of the farm's electricity came from solar

Summary

- The system produced **1,458.0 kWh** in October and the farm used **1,807 kWh**. Solar covered **77%** of the farm's electricity.
- Almost all solar was used on site (**96%**). Only about **2 kWh a day** went to the grid.
- Average output was **2.43 kWh per kWp per day**, well below what a 20 kWp array can give. The main reason: the inverter is **holding production down to match the farm's load** (export limit).

Daily production, consumption and purchases



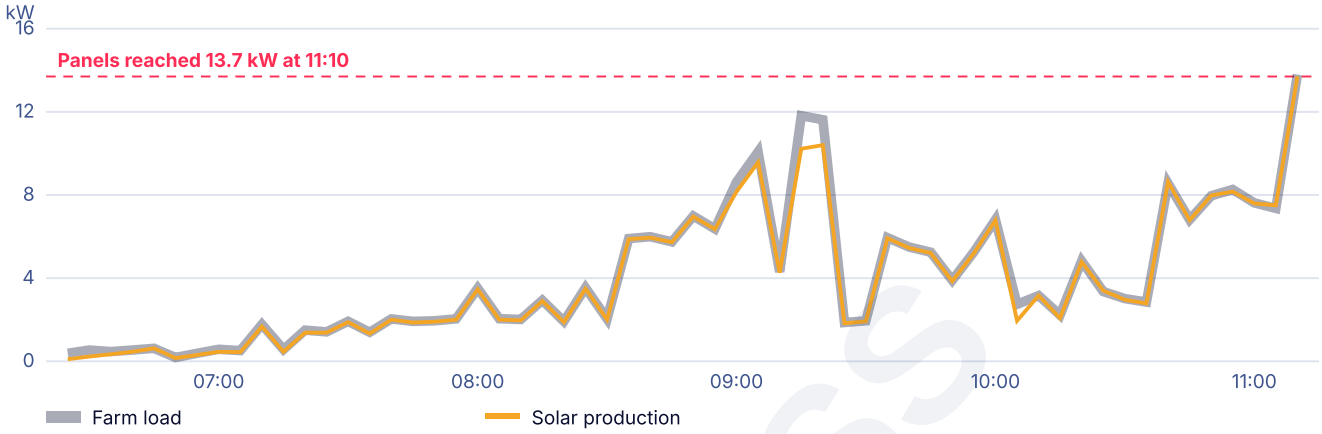
Production rises and falls with the farm's consumption rather than with the weather: on 1–2 October the farm used about 110 kWh and the panels made 89 kWh; on light-load days the panels made only 13–35 kWh.

Findings

1. **Output is capped at the farm's load.** The export limit keeps production equal to consumption. On sunny days an estimated 20–30 kWh a day of solar is not harvested.
2. **Most grid purchases are at night.** The farm buys about 14 kWh a day, mainly overnight (base load 0.2–0.5 kW with regular 0.7–1.1 kW pump or fan cycles) and in the early morning.
3. **Monitoring gaps.** Data for January, February and August is incomplete. The logger connection should be checked so no month is lost again.
4. **No alarms.** The inverter is online with no active alerts.

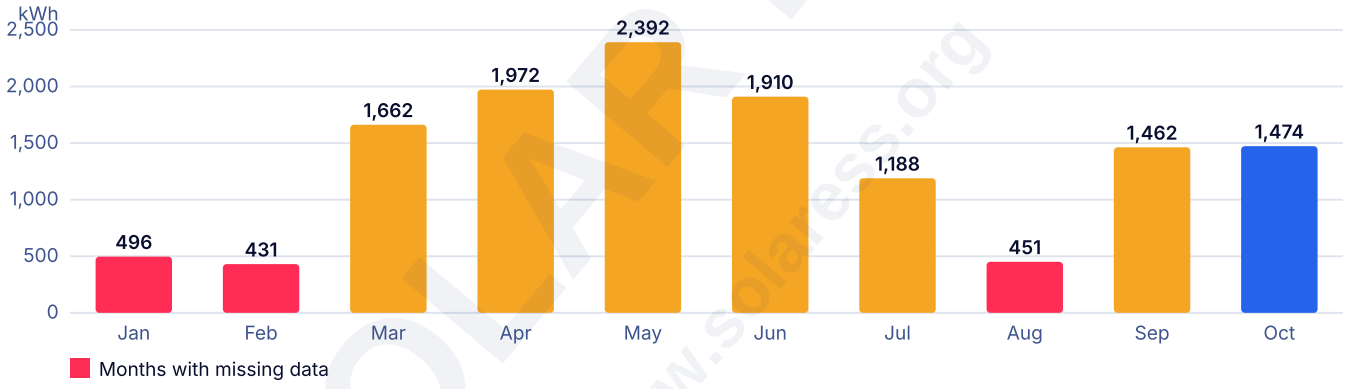
Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694 · 089 632 5829
www.solaress.org · www.โซลาร์เอสส์.com

This morning: production follows the load



31 October, 06:25–11:10, 5-minute data. The solar line sits on top of the load line all morning, within about 40 W. When the load jumped at 11:10, production rose at once to 13.7 kW: the panels can give much more than they are allowed to.

Production by month, 2026



January, February and August are far too low for this system. The inverter's own counter shows about 23,085 kWh lifetime, while the platform shows 14.61 MWh: the data logger was most likely offline in those months.

Recommendations

1. **Review the export limit.** If the farm has a sell-back agreement, raise the limit so surplus solar earns income. If not, keep the limit.
2. **Use more power in daytime.** Run pumps, fans, water heating and feed equipment between 09:00 and 15:00 where possible.
3. **Consider a battery of about 10–15 kWh.** It would store the midday solar that is now held back and cover most of the night purchases (if the inverter is a hybrid model, to be confirmed on site).
4. **Check the data logger** Wi-Fi signal and power supply.
5. **Panel cleaning** cannot be judged while output is capped: check again after the limit is changed.

Notes: figures for 1–30 October (full days). Solar used on site = production − sent to grid. Curtailment estimate based on the 13.7 kW reached at 11:10 and the farm's best months (May: 3.9 kWh/kWp/day).